Weighted Independent Sets in Latin Square Hypergraphs

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Abstract

Let D be the tripartite 3-uniform hypergraph of a Latin square of order q, and assign weight $\lambda^{|I|}$ to each independent set I. We prove, uniformly over all Latin squares, a two-term expansion with leading terms $3(1+\lambda)^{2q}$ and $3q\lambda(1+2\lambda)^q$, and an exponentially smaller error, for every fixed $0 < \lambda < (1+\sqrt{5})/2$. Elementary abelian 2-groups show that this interval is sharp, including its endpoint. We also determine the optimal uniform exponential rate of the contribution from independent sets meeting every part in at least two vertices. It is $\max\{(1+\lambda)^{3/2}, \sqrt{1+4\lambda+2\lambda^2}\}$. Finally, for Latin lifts of an arbitrary fixed 3-uniform hypergraph, we identify the largest possible exponential growth rate by a half-integral optimization on the base, and construct binary lifts attaining it.

1 Introduction

Counting independent sets in the hypergraph of a Latin square extends the problem of counting triples of subsets of a finite group that avoid a prescribed additive equation. The first contributions come from sets omitting a part and from sets meeting one part in exactly one vertex. To obtain a second-order asymptotic formula, one must control all independent sets that meet each part in at least two vertices. For arbitrary Latin squares, these remaining sets need not be concentrated near a small part. Their contribution can also come from three sets of size approximately half of a part.

A *Latin square hypergraph of order* q is a 3-uniform hypergraph D with disjoint parts X_1, X_2, X_3, each of size q, such that every edge meets each part once and every pair of vertices in distinct parts belongs to exactly one edge. A set of vertices is *independent* if it contains no edge. Write $\mathcal{I}(D)$ for the family of independent sets, and, for a fixed real number $\lambda > 0$, define

$$Z_D(\lambda) = \sum_{I \in \mathcal{I}(D)} \lambda^{|I|}.$$

Thus $Z_D(1) = |\mathcal{I}(D)|$. All independent sets in this paper have this usual meaning; they may contain two vertices of an edge.

For a finite abelian group G, the hypergraph on three copies of G whose edges satisfy $x + y = z$ is a Latin square hypergraph. Semchankau, Shabanov, and Shkredov [6] proved the unweighted expansion $3 \cdot 4^q + 3q \cdot 3^q + O((3-c)^q)$ in this setting. Araujo, Balogh, and Garcia [1, Theorem 6] sharpened the error for general finite abelian groups and obtained more detailed bounds for prime cyclic groups. Their proof uses link graphs and hypergraph containers, and their Remark 8 identifies the contribution from index-two subgroups in elementary abelian 2-groups. We use these methods and constructions for general Latin squares, keeping track of the activity λ.

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Our first result gives the range of activities for which the two elementary contributions yield a uniform two-term expansion. Put $\phi = (1+\sqrt{5})/2$.

Theorem 1.1. *For every fixed $0 < \lambda < \phi$, there is $c_\lambda > 0$ such that, uniformly over all Latin square hypergraphs D of order q,*

$$Z_D(\lambda) = 3(1+\lambda)^{2q} + 3q\lambda(1+2\lambda)^q + O_\lambda\big((1+2\lambda)^q e^{-c_\lambda q}\big) \qquad (q \to \infty). \tag{1}$$

For each fixed $\lambda \geq \phi$, no positive choice of c_λ makes (1) valid uniformly over all Latin squares. This failure already occurs for the addition tables of $\mathbb{F}_2^m$, with $q = 2^m$.

The endpoint concerns the validity of the particular expansion (1). It does not assert a common phase transition for every Latin square, or determine a complete expansion for $\lambda \geq \phi$. At $\lambda = 1$, the theorem extends the group result to arbitrary Latin squares; for example, its proof gives the uniform bound

$$|\mathcal{I}(D)| = 3 \cdot 4^q + 3q \cdot 3^q + O(2.9^q).$$

The remaining contribution has an optimal exponential rate for every fixed positive activity. Define

$$R_D(\lambda) = \sum_{\substack{I \in \mathcal{I}(D) \\ |I \cap X_i| \geq 2 \ (i=1,2,3)}} \lambda^{|I|}, \qquad \rho(\lambda) = \max\{(1+\lambda)^{3/2}, \sqrt{1+4\lambda+2\lambda^2}\}.$$

Theorem 1.2. *For every fixed $\lambda > 0$ and every $\varepsilon > 0$, uniformly over all Latin square hypergraphs D of order q,*

$$R_D(\lambda) = O_{\lambda,\varepsilon}\big((\rho(\lambda)+\varepsilon)^q\big).$$

Let D_m be the Latin square hypergraph of addition in $\mathbb{F}_2^m$. Then

$$\lim_{m\to\infty} R_{D_m}(\lambda)^{1/2^m} = \rho(\lambda).$$

This determines an exponential rate, not a polynomial factor. In particular, the asserted upper bound has $\rho(\lambda)+\varepsilon$ in place of $\rho(\lambda)$. The two expressions defining ρ agree at $(\sqrt{5}-1)/2$, whereas the cutoff in Theorem 1.1 is $(1+\sqrt{5})/2$. The distinction follows from comparing the residual contribution with the singleton contribution, whose exponential base is $1+2\lambda$.

Arras, Garbe, and Joos [2] obtain asymptotic counting formulas for regular partite hypergraphs satisfying expansion assumptions, and discuss weighted extensions. Their assumptions do not cover all Latin square hypergraphs. Indeed, the neighbourhood of any nonempty set in one part of a Latin square hypergraph consists of both other parts. For a two-vertex set this has size $2q$, which fails their condition $\mathrm{Exp}_1(\alpha)$ when $0 < \alpha < 1$.

We also consider a fixed 3-uniform hypergraph whose vertices are replaced by equal fibres and whose edges are replaced by arbitrary Latin square hypergraphs. In Section 4, we determine the largest possible exponential rate of Z over such lifts. The answer can exceed the exponent obtained by using only independent supports in the base. The affine plane of order three gives a concrete instance; its binary lifts also admit an exact classification of their largest independent sets.

The Latin lifting construction and local container estimates are also used in [4], which gives counterexamples to conjectures of Cohen, Perkins, Sarantis, and Tetali in all uniformities, together with a precise leading coefficient for lifts of regular bases attaining the incidence bound. Our main results concern sharp weighted residual rates, the endpoint of the two-term expansion, and the optimal exponential rate for arbitrary fixed triple-system bases.

Section 2 proves the uniform upper bound in Theorem 1.2. Section 3 proves the matching lower bounds and Theorem 1.1. The proofs use the container theorem of Saxton and Thomason [5], together with elementary weighted counting in unions of two perfect matchings. Throughout, logarithms are natural and all activities are fixed independently of q.

2 Uniform counting bounds

A small induced edge count forces either a small part or a total size close to $3q/2$. In the first case, two selected vertices in the small part give a useful restriction on the other two parts. We first record the container and link estimates needed for this division.

Lemma 2.1. *For every fixed $0 < \delta < 1/2$, every Latin square hypergraph D of order q has a family $\mathcal{C}$ of vertex sets such that every independent set is contained in a member of $\mathcal{C}$, each $C \in \mathcal{C}$ spans at most δq^2 edges, and*

$$\log |\mathcal{C}| = O_\delta(\sqrt{q}\log(2q)).$$

The constant is independent of the Latin square.

Proof. We apply Saxton and Thomason [5, Corollary 3.6]. The hypergraph D has $3q$ vertices, average degree q, and every set of two or three vertices belongs to at most one edge. The codegree function in their Definition 3.2 is therefore

$$\delta(D,\tau) = \frac{4}{q\tau} + \frac{2}{q\tau^2}.$$

Choose $\tau = Kq^{-1/2}$, first fixing K sufficiently large in terms of δ, and then taking q sufficiently large. This ensures $\delta(D,\tau) \le \delta/(12 \cdot 3!)$ and $\tau < 1/2$. The cited corollary gives containers with at most $\delta e(D) = \delta q^2$ edges and $\log |\mathcal{C}| = O_\delta(q\tau \log(1/\tau))$. These estimates depend only on q and δ, which proves the stated uniformity. The finitely many smaller orders can be included by increasing the implicit constant. □

For $C \subseteq V(D)$, put $c_i = |C \cap X_i|$, and write $t_+ = \max\{t, 0\}$ for a real number t. The Latin square property gives

$$e(D[C]) \ge c_i(c_j + c_k - q)_+ \qquad (\{i,j,k\} = \{1,2,3\}). \tag{2}$$

To see this, fix a vertex in $C \cap X_i$. Its link is a perfect matching between X_j and X_k, at least $c_j + c_k - q$ of whose edges have both endpoints in C. Summing over the c_i choices counts each induced edge exactly once.

The next estimate controls independent sets containing two specified vertices in the same part. For an ordinary graph G, use $Z_G(\lambda)$ for its weighted independent-set count as well.

Lemma 2.2. *Let u, v be distinct vertices of one part of D. The union G of their links on the other two parts satisfies*

$$Z_G(\lambda) \le (1 + 4\lambda + 2\lambda^2)^{q/2}.$$

Proof. Each link is a perfect matching. The two matchings have no common edge, because a pair in different parts belongs to only one hyperedge. Consequently, G is a disjoint union of simple even cycles $C_{2\ell}$, where $\ell \ge 2$ and the values of ℓ sum to q.

The transfer matrix

$$T = \begin{pmatrix} 1 & \lambda \\ 1 & 0 \end{pmatrix}$$

counts the choices on a cycle by $Z_{C_s}(\lambda) = \operatorname{tr}(T^s)$. Its two eigenvalues are

$$u_\lambda = \frac{1 + \sqrt{1+4\lambda}}{2}, \qquad v_\lambda = \frac{1 - \sqrt{1+4\lambda}}{2}.$$

For $x = u_\lambda^2$ and $y = v_\lambda^2$, both nonnegative,

$$Z_{C_{2\ell}}(\lambda) = x^\ell + y^\ell \le (x^2 + y^2)^{\ell/2} = (1 + 4\lambda + 2\lambda^2)^{\ell/2}.$$

The inequality follows by normalizing $x^2 + y^2 = 1$ and using $x^\ell \le x^2$ and $y^\ell \le y^2$ for $\ell \ge 2$. Multiplication over the disjoint cycles proves the assertion. □

We can now bound all residual configurations. Fixing the small parameters before letting q tend to infinity is essential for a uniform estimate.

Proof of the upper bound in Theorem 1.2. Abbreviate

$$a = 1 + \lambda, \qquad h = 1 + 4\lambda + 2\lambda^2.$$

Fix $0 < \gamma < 1/2$, and apply Lemma 2.1 with $\delta = \gamma^2$. Let C be one of its containers.

Suppose first that $c_i \le \gamma q$ for some i. Every independent set counted by $R_D(\lambda)$ and contained in C has a pair of selected vertices in $C \cap X_i$. Sum over all choices of this pair. Its weight is λ^2; the remaining choices in $C \cap X_i$ have total weight at most $a^{\gamma q}$. By Lemma 2.2, the choices in the other two parts have total weight at most $h^{q/2}$. A set can be counted more than once in this sum, which is harmless for an upper bound. Thus the contribution from C is at most $q^2\lambda^2 a^{\gamma q} h^{q/2}$.

Suppose instead that $c_i > \gamma q$ for all i. Since $e(D[C]) \le \gamma^2 q^2$, inequality (2) implies $c_j + c_k \le (1+\gamma)q$ for each pair of parts. Adding the three inequalities gives $|C| \le (3/2+3\gamma/2)q$. The weight of all subsets of C is therefore at most $a^{(3/2+3\gamma/2)q}$.

The containers cover every independent set, so

$$R_D(\lambda) \le \exp\bigl(O_\gamma(\sqrt{q}\log q)\bigr)\left(q^2\lambda^2 a^{\gamma q} h^{q/2} + a^{(3/2+3\gamma/2)q}\right). \tag{3}$$

Given $\varepsilon > 0$, choose $\gamma > 0$ so small that both $a^\gamma\sqrt{h}$ and $a^{3/2+3\gamma/2}$ are strictly smaller than $\rho(\lambda) + \varepsilon$. For this fixed γ, the exponential factor and the polynomial factor in (3) can be absorbed into the remaining strict gap for sufficiently large q. This proves the claimed bound with constants depending only on λ and ε. □

3 The expansion and its sharpness

The contributions from an empty part or a singleton part have exact formulas independent of the Latin square. We compute these first. The residual bound then proves the expansion, while binary group examples show why its range cannot be enlarged.

Fix $\lambda > 0$, and retain the notation

$$a = 1 + \lambda, \qquad b = 1 + 2\lambda, \qquad h = 1 + 4\lambda + 2\lambda^2.$$

Let U_0 be the total weight of independent sets omitting at least one part, and let U_1 be the total weight of independent sets meeting all three parts and meeting at least one part in exactly one vertex. These two families and the family counted by $R_D(\lambda)$ form a disjoint partition of $\mathcal{I}(D)$.

Lemma 3.1. *For every Latin square hypergraph of order q,*

$$\begin{aligned} U_0 &= 3a^{2q} - 3a^q + 1, \\ U_1 &= 3q\lambda(b^q - 2a^q + 1) - 3q^2\lambda^2(a^{q-1} - 1) + (q^3 - q^2)\lambda^3. \end{aligned}$$

Consequently,

$$Z_D(\lambda) = 3a^{2q} + 3q\lambda b^q + R_D(\lambda) + E_q(\lambda), \qquad E_q(\lambda) = O_\lambda(q^2 a^q + q^3). \tag{4}$$

Proof. If a specified part is empty, every subset of the other two parts is independent and has total weight a^{2q}. Intersections of two such events have weight a^q, and their triple intersection consists only of the empty set. Inclusion–exclusion gives U_0.

For U_1, first specify the singleton part and its selected vertex. The other two parts must form an independent set in a perfect matching, whose weighted count is b^q. Requiring both parts to be nonempty changes this count to $b^q - 2a^q + 1$. There are three choices of the part and q choices of the vertex, whose weight is λ.

If two specified parts are singletons, their two vertices exclude exactly one vertex of the third part. The nonempty choices in that part have total weight $a^{q-1} - 1$. This accounts for the subtraction $3q^2\lambda^2(a^{q-1} - 1)$. Finally, if every part is a singleton, exactly $q^3 - q^2$ triples are nonedges. Inclusion–exclusion gives the formula for U_1. After subtracting $3a^{2q} + 3q\lambda b^q$ from $U_0 + U_1$, the remaining terms are

$$-3a^q + 1 - 6q\lambda a^q + 3q\lambda - 3q^2\lambda^2 a^{q-1} + 3q^2\lambda^2 + (q^3 - q^2)\lambda^3.$$

They satisfy the estimate in (4). □

Proof of Theorem 1.1 for $0 < \lambda < \phi$. The identities

$$b^2 - h = 2\lambda^2 > 0, \qquad b^2 - a^3 = \lambda(1 + \lambda - \lambda^2)$$

show that $\rho(\lambda) < b$ precisely when $0 < \lambda < \phi$. Choose $\varepsilon > 0$ so that $\rho(\lambda) + \varepsilon < b$. The already proved upper bound in Theorem 1.2 makes $R_D(\lambda)$ exponentially smaller than b^q, uniformly in D. Since $1 < a < b$, the same is true of $E_q(\lambda)$ in (4). Substitution into that identity proves (1). For $\lambda = 1$, one has $\rho(1) = 2^{3/2} < 2.9$. Taking $\varepsilon = 2.9 - 2^{3/2}$ in Theorem 1.2 also gives the stated error $O(2.9^q)$. □

It remains to prove sharpness and the lower bound in Theorem 1.2. Both come from addition in $\mathbb{F}_2^m$, but use different families of independent sets. The following halfspace construction is the weighted version of the index-two subgroup construction discussed in [1, Remark 8].

Lemma 3.2. *Let D_m be the addition hypergraph of $G = \mathbb{F}_2^m$, and put $q = 2^m$. For every fixed $\lambda > 0$,*

$$R_{D_m}(\lambda) \geq (1 - o_\lambda(1))qa^{3q/2} \qquad (m \to \infty). \tag{5}$$

For every $m \geq 1$, one also has

$$R_{D_m}(\lambda) \geq \lambda^2 \left(h^{q/2} - 2a^q - 2q\lambda a^{q-2}\right). \tag{6}$$

Proof. For each nonzero linear map $f : G \to \mathbb{F}_2$, let $B_f = \{x \in G : f(x) = 1\}$. The set B_f has size $q/2$. Selecting vertices from B_f in all three parts gives an independent set, since $1 + 1 \neq 1$ in $\mathbb{F}_2$. The total weight of these choices with at least two selected vertices in each part is

$$W_q = \left(a^{q/2} - 1 - (q/2)\lambda\right)^3.$$

There are $q - 1$ nonzero maps. For distinct f, g, the maps are linearly independent, so $|B_f \cap B_g| = q/4$. The weight of the intersection of their two families is at most $a^{3q/4}$. The first two terms of inclusion–exclusion therefore give

$$R_{D_m}(\lambda) \geq (q-1)W_q - \binom{q-1}{2} a^{3q/4} = (1 - o_\lambda(1))qa^{3q/2}.$$

Here $m \geq 2$ suffices for the intersection calculation, and only the limit is used.

For the second bound, select exactly 0 and a fixed nonzero $u \in G$ in the first part. Their two links have edges $z = y$ and $z = y + u$. Since $u + u = 0$, the union consists of $q/2$ disjoint 4-cycles. Its independent-set weight is exactly $h^{q/2}$. We must exclude choices containing at

most one vertex in either remaining part. If one specified part is empty, the weight is a^q. If it is a singleton, its selected vertex has two distinct neighbours, and the weight summed over all singleton choices is $q\lambda a^{q-2}$. The union bound over the two parts removes weight at most $2a^q + 2q\lambda a^{q-2}$. Multiplying by the weight λ^2 of the selected pair proves (6). □

Completion of Theorem 1.2. The first bound in Lemma 3.2 gives

$$\liminf_{m\to\infty} R_{D_m}(\lambda)^{1/2^m} \geq a^{3/2}.$$

Since $h - a^2 = 2\lambda + \lambda^2 > 0$, the subtracted terms in (6) are $o_\lambda(h^{q/2})$. That bound gives the additional lower estimate $\sqrt{h}$. Their maximum is $\rho(\lambda)$, while the uniform upper bound in Theorem 1.2, with any $\varepsilon > 0$, gives limsup at most $\rho(\lambda) + \varepsilon$. Letting ε decrease to zero proves the limit. □

Completion of Theorem 1.1. For $\lambda \geq \phi$, one has $a^{3/2} \geq b$. Equations (4) and (5), together with $q^2a^q + q^3 = o_\lambda(qa^{3q/2})$, imply along $q = 2^m$ that

$$Z_{D_m}(\lambda) - 3a^{2q} - 3q\lambda b^q \geq (1 - o_\lambda(1))qa^{3q/2} \geq (1 - o_\lambda(1))qb^q.$$

This contradicts an error $O_\lambda(b^q e^{-cq})$ for any $c > 0$. The argument includes equality $\lambda = \phi$. □

For completeness, the identity $a^3 - h = \lambda(\lambda^2 + \lambda - 1)$ explains the internal change in the residual rate. Below $(\sqrt{5} - 1)/2$, the two-link construction has the larger exponential rate; above it, the halfspace construction does. Neither observation determines the polynomial factor in $R_{D_m}(\lambda)$, or the residual behaviour of every individual sequence of Latin squares.

4 Lifts of a fixed hypergraph

The local estimates also identify the largest possible counting exponent when several Latin square hypergraphs share fibres. In this setting, a family of independent sets can be supported on vertices of the base that themselves contain edges. Synchronized halfspaces give such families and can dominate all families supported on independent sets of the base.

Let A be a fixed finite simple 3-uniform hypergraph with N vertices. For each $v \in V(A)$, let V_v be a fibre of q vertices. For each edge $uvw \in E(A)$, place an arbitrary Latin square hypergraph of order q on V_u, V_v, V_w. Denote the union by $L_q(A)$. The Latin squares assigned to different edges may be different. This definition does not require A to be regular or linear. If A is k-regular and linear, then $L_q(A)$ is kq-regular and linear.

For $S \subseteq V(A)$, let $\mathrm{iso}_A(S)$ be the number of isolated vertices in the induced hypergraph $A[S]$, and define

$$\beta(A) = \max_{S\subseteq V(A)} \frac{|S| + \mathrm{iso}_A(S)}{2}. \tag{7}$$

Isolation here refers to $A[S]$, not to A. Write $\alpha(A)$ for the largest size of an independent set in A. An independent set S has $\mathrm{iso}_A(S) = |S|$, so $\beta(A) \geq \alpha(A)$. Also $\beta(A) \geq N/2$.

Theorem 4.1. *For every fixed finite simple 3-uniform hypergraph A, the following statements hold.*

1. *Every Latin lift satisfies $\alpha(L_q(A)) \leq \beta(A)q$.*
2. *For every fixed $\lambda > 0$ and $\eta > 0$, all sufficiently large q satisfy*

$$Z_{L_q(A)}(\lambda) \leq (1 + \lambda)^{(\beta(A)+\eta)q},$$

uniformly over the Latin squares assigned to the base edges.

3. *For $q = 2^m$ with $m \geq 1$, label every fibre by $\mathbb{F}_2^m$ and use $x + y + z = 0$ on every base edge. Denote this synchronized lift by $L_q^{\mathrm{bin}}(A)$. Then*

$$\alpha(L_q^{\mathrm{bin}}(A)) = \beta(A)q, \qquad \lim_{m\to\infty} \frac{\log Z_{L_{2^m}^{\mathrm{bin}}(A)}(\lambda)}{2^m \log(1+\lambda)} = \beta(A).$$

The upper bound concerns all choices of local Latin squares. The matching lower bound uses a particular synchronized family; it does not assert that every choice attains this exponent. The base A is fixed throughout the limit. No leading coefficient or second term is asserted for these lifts.

The optimization in (7) follows from the half-integrality of the fractional independence polytope of an ordinary graph. We include the short argument, a standard fact also recorded in [3, Lemma 12].

Lemma 4.2. *For a graph G, put*

$$f(G) = \max\left\{ \sum_{v\in V(G)} x_v : 0 \leq x_v \leq 1, \quad x_u + x_v \leq 1 \text{ for } uv \in E(G) \right\}.$$

There is an optimal vector with every coordinate in $\{0, \frac{1}{2}, 1\}$. For $S \subseteq V(A)$, let G_S be the graph on S in which uv is an edge if $uvw \in E(A[S])$ for some $w \in S$. Then

$$\max_{S\subseteq V(A)} f(G_S) = \beta(A). \tag{8}$$

Proof. Choose an extreme optimal vector for the polytope defining $f(G)$. Consider the coordinates strictly between zero and one, and the graph of tight constraints $x_u + x_v = 1$ between these coordinates. If a connected component of this graph is bipartite, increase the coordinates on one side and decrease those on the other by a small common amount. Both this perturbation and its negative preserve the tight constraints. No tight edge joins an interior coordinate to a coordinate equal to zero or one, and every other constraint has positive slack. A sufficiently small perturbation is therefore feasible in both directions, contradicting extremality. Each component must instead contain an odd cycle, which forces every coordinate in that component to equal $1/2$.

For any S, assigning 1 to the isolated vertices of $A[S]$ and $1/2$ to its other vertices is feasible for G_S. This proves that the left side of (8) is at least $\beta(A)$.

Conversely, take a half-integral optimal vector for G_S, and let S' be the set of its positive coordinates. A coordinate equal to 1 has no neighbour of positive weight in G_S; its vertex is therefore isolated in $A[S']$. Raise every other isolated coordinate in $A[S']$ to 1, and keep the remaining coordinates at $1/2$. The resulting sum is $(|S'| + \mathrm{iso}_A(S'))/2$, at least the original value $f(G_S)$ and at most $\beta(A)$. This argument uses only that $G_{S'}$ is a subgraph of $G_S[S']$; these two graphs need not be equal. Equation (8) follows. □

Proof of Theorem 4.1. Let I be an independent set in $L_q(A)$, and let $S = \{v : I \cap V_v \neq \varnothing\}$. Set $x_v = |I \cap V_v|/q$ for $v \in S$. If $uvw \in E(A[S])$, fix a selected vertex in V_w. Its link is a perfect matching between V_u and V_v. Since I contains no complete edge, it selects at most one endpoint from each matching edge, giving $x_u + x_v \leq 1$. The same reasoning applies to each pair of the base edge. Thus x is feasible for G_S, and

$$|I|/q \leq f(G_S) \leq \beta(A)$$

by Lemma 4.2. This proves the first assertion.

For the counting upper bound, fix $0 < \gamma < 1/2$. Apply Lemma 2.1, with $\delta = \gamma^2$, to the local Latin square on each base edge. For each tuple of local containers, intersect their vertex sets within each common fibre. Fibres of isolated base vertices remain unrestricted. The resulting global containers cover all independent sets of $L_q(A)$. Since the number of base edges is fixed, their number is $\exp(O_{A,\gamma}(\sqrt{q}\log q))$.

For one global container C, put $c_v = |C \cap V_v|$ and $S = \{v : c_v \geq \gamma q\}$. On each base edge $uvw \subseteq S$, the local form of (2) gives $c_w(c_u + c_v - q)_+ \leq \gamma^2 q^2$, hence $c_u + c_v \leq (1+\gamma)q$. Therefore the vector $c_v/((1+\gamma)q)$, $v \in S$, is feasible for G_S. The fibres outside S each contribute fewer than γq vertices. Lemma 4.2 yields

$$|C| \leq q(1+\gamma)\beta(A) + N\gamma q.$$

The weight of all subsets of C is $(1+\lambda)^{|C|}$. Summing over the global containers gives

$$Z_{L_q(A)}(\lambda) \leq \exp(O_{A,\gamma}(\sqrt{q}\log q))(1+\lambda)^{[\beta(A)+\gamma(\beta(A)+N)]q}.$$

Given $\eta > 0$, first choose γ with $\gamma(\beta(A)+N) < \eta/2$. Then take q sufficiently large to absorb the container count into $(1+\lambda)^{\eta q/2}$. This proves the second assertion uniformly over the local designs.

Finally, choose S attaining (7). Let T be the isolated vertices of $A[S]$, and fix a nonzero linear map $f : \mathbb{F}_2^m \to \mathbb{F}_2$. In the synchronized binary lift, take the entire fibre at each $v \in T$, the halfspace $f^{-1}(1)$ at each $v \in S \setminus T$, and no vertices in the other fibres. The resulting set W has size

$$|W| = q|T| + (q/2)|S \setminus T| = \beta(A)q.$$

A base edge not contained in S misses a selected fibre. A base edge contained in S avoids T, so the three selected labels all have f-value 1; their sum cannot be zero. Thus W is independent. The first assertion gives equality for the independence number, and all subsets of W give $Z_{L_q^{\mathrm{bin}}(A)}(\lambda) \geq (1+\lambda)^{\beta(A)q}$. Together with the second assertion for every $\eta > 0$, this proves the limiting formula. □

The next example shows why the support of an independent set in a lift need not be independent in the base. It also illustrates how the binary construction can determine the largest sets exactly.

Proposition 4.3. *Let A be the 3-uniform hypergraph of affine lines in $\mathbb{F}_3^2$. Then $\alpha(A) = 4$ and $\beta(A) = 9/2$. For $q = 2^m$ with $m \geq 1$, the synchronized binary lift $H_q = L_q^{\mathrm{bin}}(A)$ is a linear $4q$-regular hypergraph on $9q$ vertices satisfying*

$$\alpha(H_q) = 9q/2, \qquad \log Z_{H_q}(\lambda) = (9/2 + o_\lambda(1))q\log(1+\lambda).$$

It has exactly $q-1$ largest independent sets. They are obtained by choosing a nonzero linear map $f : \mathbb{F}_2^m \to \mathbb{F}_2$ and selecting $f^{-1}(1)$ in every fibre.

Proof. The four points $\{(0,0),(0,1),(1,0),(1,1)\}$ contain no affine line. If a set of five points contained no line, each of its ten pairs would complete to a line at a point in its four-point complement. For a fixed complement point, the four lines through it partition the other eight points into pairs, so at most two of the ten pairs could complete there. This would account for at most eight pairs, a contradiction. Hence $\alpha(A) = 4$.

For $|S| \geq 6$, no vertex of $A[S]$ is isolated. Indeed, the four lines through one vertex are disjoint outside that vertex, and deleting at most three points cannot meet all four. Thus $(|S| + \mathrm{iso}_A(S))/2 = |S|/2$ in this range. If $|S| = 5$, it contains a line, so $\mathrm{iso}_A(S) \leq 2$ and its value is at most $7/2$. If $|S| \leq 4$, its value is at most 4. It follows that $\beta(A) = 9/2$, attained only by $S = V(A)$. Theorem 4.1 proves the numerical assertions about H_q.

To classify its largest independent sets, first observe from the proof of Lemma 4.2 that $f(G_S) \leq 4$ for every proper subset $S \subsetneq V(A)$. An independent set I of size $9q/2$ must therefore meet every fibre. Every pair of base vertices belongs to a line, so $|I \cap V_u| + |I \cap V_v| \leq q$ for all distinct u, v. Summing these 36 inequalities counts each fibre size eight times. The sum equals $36q$, and hence every inequality is an equality. All nine fibre intersections have size $q/2$.

We need one elementary fact about half-size subsets of $G = \mathbb{F}_2^m$. Suppose $X, Y, Z \subseteq G$ all have size $q/2$ and $x + y + z = 0$ has no solution in $X \times Y \times Z$. For each $z \in Z$, the disjointness condition gives $X + z = G \setminus Y$. Fix $z_0 \in Z$. Then $Z + z_0$ lies in the translation stabilizer $K = \{g \in G : X + g = X\}$. This is a proper subgroup, because X is nonempty and not all of G. Since $|K| \geq |Z| = q/2$, it has index two. The sets X, Y, Z are consequently cosets of the same subgroup K.

Apply this fact to each affine line of the base. A half-size coset has a unique translation stabilizer, and the affine plane is connected through its lines. Thus all nine fibres use cosets of one common index-two subgroup $\ker f$, for a nonzero linear map $f : G \to \mathbb{F}_2$. Write $s_v \in \mathbb{F}_2$ for the coset selected at the base vertex v. The independence condition is $s_u + s_v + s_w = 1$ on every affine line uvw. Set $y_v = s_v + 1$. Each line then has y-sum zero. Summing over a parallel class gives $\sum_v y_v = 0$. Summing over the four lines through a fixed vertex v gives $\sum_{u \neq v} y_u = 0$, since v appears four times. Therefore $y_v = 0$ for every v, and all $s_v = 1$. Each of the $q - 1$ nonzero maps f gives a distinct set, as claimed. □

The distinction from counting only independent supports can be made explicit. Put $P_A(t) = \sum_{S \in \mathcal{I}(A)} t^{|S|}$. In any lift, the independent sets whose nonempty fibres form an independent set of the base have total weight

$$P_A\big((1 + \lambda)^q - 1\big).$$

For the affine plane this is $(1 + \lambda)^{(4+o_\lambda(1))q}$, strictly below the rate $(1 + \lambda)^{(9/2+o_\lambda(1))q}$ in Proposition 4.3. Thus those supports carry an exponentially small proportion of the total weight in the binary lift.

For regular bases, the possible gain remains constrained by the usual degree bound on the independence number.

Corollary 4.4. *If A is k-regular with $k \geq 1$, then*

$$\beta(A) \leq \frac{2N}{3}, \qquad \beta(A) = \frac{2N}{3} \quad \Longleftrightarrow \quad \alpha(A) = \frac{2N}{3}.$$

Proof. Choose S attaining (7), and assign weight 1 to the isolated vertices of $A[S]$, weight $1/2$ to its other vertices, and weight zero outside S. The weights sum to $\beta(A)$. A base edge contained in S has total weight $3/2$, while an edge not contained in S has total weight at most 2. Regularity therefore gives

$$k\beta(A) = \sum_{e \in E(A)} \sum_{v \in e} x_v \leq 2|E(A)| = \frac{2Nk}{3}.$$

Equality forces every base edge to have weights $1, 1, 0$. Since $k \geq 1$, no vertex then has weight $1/2$, and the positive support is an independent set of size $2N/3$. Conversely, an independent set of this size contributes $2N/3$ to the maximum in (7). □

In particular, if a fixed regular base satisfies $\alpha(A) < 2N/3$, every sequence of its Latin lifts satisfies

$$\limsup_{q \to \infty} \frac{\log_2 Z_{L_q(A)}(1)}{Nq} \leq \frac{\beta(A)}{N} < \frac{2}{3}.$$

Theorem 4.1 identifies the optimal exponent within the fixed-base model; it does not give a general upper bound for all regular linear hypergraphs. The polynomial factors for binary lifts, and the full expansion for a single Latin square at activities $\lambda \geq \phi$, remain open.

Declaration of AI use

The author used GPT-6.1 to assist with exploring proof strategies, checking calculations, and preparing the exposition. The author takes responsibility for the content of this paper.

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